TIKTOK DATA ANALYSIS

datalab

TIKTOK DATA ANALYSIS PROJECT

```python
# Import packages
import pandas as pd
import numpy as np

# Load dataset into dataframe
data = pd.read_csv("tiktok_dataset.csv")

data.head(10)
data.info()
data.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 19382 entries, 0 to 19381
Data columns (total 12 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   #                         19382 non-null  int64
 1   claim_status              19084 non-null  object
 2   video_id                  19382 non-null  int64
 3   video_duration_sec        19382 non-null  int64
 4   video_transcription_text  19084 non-null  object
 5   verified_status           19382 non-null  object
 6   author_ban_status         19382 non-null  object
 7   video_view_count          19084 non-null  float64
 8   video_like_count          19084 non-null  float64
 9   video_share_count         19084 non-null  float64
 10  video_download_count      19084 non-null  float64
 11  video_comment_count       19084 non-null  float64
dtypes: float64(5), int64(3), object(4)
memory usage: 1.8+ MB
```

	#	video_id	video_duration...	video_view_c...	video_like_co...	video_share_...	video_download_c...	video_comment_...
count	19382	19382	19382	19084	19084	19084	19084	19084
mean	9691.5	5627454067.339129	32.4217315035	254708.5586879061	84304.6360301824	16735.2483232027	1049.4296269126	349.3121463006
std	5595.2457944223	2536440464.1693673	16.2299671785	322893.2808137871	133420.5468136723	32036.1743501506	2004.2998939792	799.6388647869
min	1	1234959018	5	20	0	0	0	0
25%	4846.25	3430416807.25	18	4942.5	810.75	115	7	1
50%	9691.5	5618663579	32	9954.5	3403.5	717	46	9
75%	14536.75	7843960211.25	47	504327	125020	18222	1156.25	292
max	19382	9999873075	60	999817	657830	256130	14994	9599

Rows: 8 Expand

```
# What are the different values for claim status and how many of each are in the data?
claimstatus_cnt = data.groupby('claim_status').size()
print(claimstatus_cnt)
```

```
claim_status
claim      9608
opinion    9476
dtype: int64
```

```
# What is the average view count of videos with "claim" status?
claim = data['claim_status'] == 'claim'
cnt_claim = data[claim]['video_view_count'].mean()
print('Mean view count claims:',cnt_claim)
```

```
Mean view count claims: 501029.4527477102
```

```
# What is the average view count of videos with "opinion" status?
opinion = data['claim_status'] == 'opinion'
cnt_opinion = data[opinion]['video_view_count'].mean()
print('Median view count claims:',cnt_opinion)
```

```
Median view count claims: 4956.43224989447
```

```
# Get counts for each group combination of claim status and author ban status
data.groupby(['claim_status', 'author_ban_status']).count()[['#']]
```

claim_status	author_ban_status	#
claim	active	6566
claim	banned	1439
claim	under review	1603
opinion	active	8817
opinion	banned	196
opinion	under review	463

Rows: 6 Expand

```
auth_ban_stat = data.groupby(['author_ban_status']).median(numeric_only=True)[
    ['video_share_count']]
# What's the median video share count of each author ban status?
auth_ban_stat
```

author_ban_status	video_share_count
active	437
banned	14468
under review	9444

Rows: 3 Expand

```
# In depth Analysis on Author ban status
agg_video_stat = data.groupby('author_ban_status').agg({'video_view_count': ['mean', 'median'],
                                                        'video_like_count': ['mean', 'median'],
                                                        'video_share_count': ['mean', 'median']})
agg_video_stat
```

author_ban_s...	('video_view_count', 'mean')	('video_view_count', 'median')	('video_like_count', 'mean')	('video_like_count', 'median')	('video_share_count', 'mean')
	215927.03952415	8616	71036.5338360528	2222	14111.4661639472
	445845.4391437309	448201	153017.2366972477	105573	29998.9425076453
	392204.8363988383	365245.5	128718.050338819	71204.5	25774.6969990319

Rows: 3 Expand

```
# Create a likes_per_view column
data['likes_per_view'] = data['video_like_count'] / data['video_view_count']

# Create a comments_per_view column
data['comments_per_view'] = data['video_comment_count'] / data['video_view_count']

# Create a shares_per_view column
data['shares_per_view'] = data['video_share_count'] / data['video_view_count']

agg_perview_stat = data.groupby(['claim_status', 'author_ban_status']).agg({'likes_per_view': ['mean', 'median'],
                                                        'comments_per_view': ['mean', 'median'],
                                                        'shares_per_view': ['mean', 'median']})
agg_perview_stat
```

clai...	author_ban_s...	('likes_per_view', 'mean')	('likes_per_view', 'median')	('comments_per_view', 'mean')	('comments_per_view', 'median')	('shares_per_view', 'mean')
		0.3295420302	0.326538133	0.0013926289	0.0007755095	0.0654557881
		0.3450705546	0.3589086721	0.0013771889	0.0007456234	0.067893415
		0.3279966584	0.3208674167	0.0013670879	0.0007888666	0.0657334689
		0.2197435214	0.2183303085	0.0005170828	0.0002517623	0.0437285744
		0.2068684015	0.1984830353	0.0004343612	0.0001927651	0.0405307243
		0.2263943726	0.2280508007	0.0005355554	0.0002926116	0.0444724222

Rows: 6 Expand